CLASS 10 — MATHS · FORMULA SHEET

Chapter 11: **Areas Related to Circles** | Quick revision, one page

Circle

- Circumference $= 2\pi r = \pi d$. Area $= \pi r^2$. Take $\pi = \dfrac{22}{7}$ unless told otherwise.
- Ring (annulus) between radii R and r: area $= \pi(R^2 - r^2) = \pi(R + r)(R - r)$.

Sector of angle θ (in degrees)

- Length of arc $= \dfrac{\theta}{360^\circ} \times 2\pi r$.
- Area of sector $= \dfrac{\theta}{360^\circ} \times \pi r^2 = \dfrac{1}{2}\, l\, r$ (l = arc length).
- Perimeter of sector $= 2r + \text{arc length}$.
- Quadrant: $\theta = 90^\circ$. Semicircle: $\theta = 180^\circ$.

Segment of a circle

- Area of minor segment = area of sector − area of $\triangle OAB$.
- Area of major segment $= \pi r^2 -$ area of minor segment.
- $\triangle OAB$: $\theta = 90^\circ$: $\frac{1}{2}r^2$; $\theta = 60^\circ$: $\frac{\sqrt{3}}{4}r^2$; $\theta = 120^\circ$: $\frac{\sqrt{3}}{4}r^2$.

Clocks and wheels

- Minute hand turns 360° in 60 min (6° per minute); hour hand 360° in 12 h (0.5° per minute).
- Distance covered by a wheel in n revolutions $= n \times 2\pi r$.

Combinations

- Area of shaded region = area of larger figure − area of smaller figures (squares, triangles, circles, quadrants).

MASTERMIND